Chapter 6 • Trigonometry

Solved Exercise 6.2

1. For each right triangle, find (i) sinθ, (ii) cosθ, (iii) tanθ, (iv) secθ, (v) cosecθ, (vi) cotφ, (vii) tanφ, (viii) cosecφ, (ix) secφ and (x) cosφ.

(a) Relative to θ, perpendicular=4, base=3, hypotenuse=5; relative to φ, perpendicular and base interchange.

(i) $\sin\theta=\frac{4}{5}$

(ii) $\cos\theta=\frac{3}{5}$

(iii) $\tan\theta=\frac{4}{3}$

(iv) $\sec\theta=\frac{5}{3}$

(v) $\operatorname{cosec}\theta=\frac{5}{4}$

(vi) $\cot\varphi=\frac{4}{3}$

(vii) $\tan\varphi=\frac{3}{4}$

(viii) $\operatorname{cosec}\varphi=\frac{5}{3}$

(ix) $\sec\varphi=\frac{5}{4}$

(x) $\cos\varphi=\frac{4}{5}$

(b) Relative to θ, perpendicular=8, base=15, hypotenuse=17.

(i) $\sin\theta=\frac{8}{17}$

(ii) $\cos\theta=\frac{15}{17}$

(iii) $\tan\theta=\frac{8}{15}$

(iv) $\sec\theta=\frac{17}{15}$

(v) $\operatorname{cosec}\theta=\frac{17}{8}$

(vi) $\cot\varphi=\frac{8}{15}$

(vii) $\tan\varphi=\frac{15}{8}$

(viii) $\operatorname{cosec}\varphi=\frac{17}{15}$

(ix) $\sec\varphi=\frac{17}{8}$

(x) $\cos\varphi=\frac{8}{17}$

(c) Relative to θ, perpendicular=5, base=12, hypotenuse=13.

(i) $\sin\theta=\frac{5}{13}$

(ii) $\cos\theta=\frac{12}{13}$

(iii) $\tan\theta=\frac{5}{12}$

(iv) $\sec\theta=\frac{13}{12}$

(v) $\operatorname{cosec}\theta=\frac{13}{5}$

(vi) $\cot\varphi=\frac{5}{12}$

(vii) $\tan\varphi=\frac{12}{5}$

(viii) $\operatorname{cosec}\varphi=\frac{13}{12}$

(ix) $\sec\varphi=\frac{13}{5}$

(x) $\cos\varphi=\frac{5}{13}$

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2. In right triangle ABC, B=90°, AC=b, BC=a, AB=c, A=φ and C=θ. Find the six requested ratios.

(i) sinθ=opposite/hypotenuse=AB/AC=c/b.
(ii) cosθ=adjacent/hypotenuse=BC/AC=a/b.
(iii) tanθ = AB/BC
$= \frac{c}{a}$
(iv) sinφ = BC/AC
= a/b
(v) cosφ = AB/AC
= c/b
(vi) tanφ = BC/AB
$= \frac{a}{c}$

3. In the shown right triangle, the side opposite θ is 3, the adjacent side is 4 and the hypotenuse is 5. Verify the three reciprocal products.

(i) sinθ·cosecθ = (3/5)(5/3)
= 1.
(ii) cosθ·secθ = (4/5)(5/4)
= 1.
(iii) tanθ·cotθ = (3/4)(4/3)
= 1.

4. Complete each complementary-angle statement.

(i) sin30° = sin(90°−60°)
= cos60°
$= \frac{1}{2}$
(ii) cos30° = cos(90°−60°)
= sin60°
$= \frac{\sqrt{3}}{2}$
(iii) tan30° = tan(90°−60°)
= cot60°
$= \frac{1}{\sqrt{3}}$
(iv) tan60° = tan(90°−30°)
= cot30°
= √3.
(v) sin60° = sin(90°−30°)
= cos30°
$= \frac{\sqrt{3}}{2}$
(vi) cos60° = cos(90°−30°)
= sin30°
$= \frac{1}{2}$
(vii) sin45° = sin(90°−45°)
= cos45°
$= \frac{1}{\sqrt{2}}$
(viii) tan45° = tan(90°−45°)
= cot45°
= 1.
(ix) cos45° = cos(90°−45°)
= sin45°
$= \frac{1}{\sqrt{2}}$

5. In right triangle ABC, B=90°, C=60°, A=30° and sinA=$\frac{a}{b}$ Evaluate the ten requested ratios.

(i) mBC/mAB = a/c
= tan30°
$= \frac{1}{\sqrt{3}}$
(ii) cos60°=$\frac{1}{2}$
(iii) tan60°=√3.

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(iv) $\operatorname{cosec}(\pi/3) = \operatorname{cosec}60°$
$= \frac{2}{\sqrt{3}}$
(v) $\cot 60° = \frac{1}{\sqrt{3}}$
(vi) $\sin 30° = \frac{1}{2}$
(vii) $\cos 30° = \frac{\sqrt{3}}{2}$
(viii) $\tan(\pi/6) = \tan 30°$
$= \frac{1}{\sqrt{3}}$
(ix) $\sec 30° = \frac{2}{\sqrt{3}}$
(x) $\cot 30° = \sqrt{3}$.

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